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Boston University
M.S. Robotics and Autonomous System (expected 2025)

GPA 3.73/4
Rochester Institute of Technology
B.S. Computer Science
GPA 3.85/4
Expected Graduation: May 2022

Employment

- D3 Engineering** · Engineering Technician Jan. 2021 to Dec. 2021
- Developed Linux Kernel drivers for cameras designed for embedded vision on the NVIDIA Jetson platform
 - Optimized for low latency camera frame throughput
 - Populated Linux device trees for newly designed hardware
 - Added features to NVIDIA Linux Kernel to expand camera platform capabilities
 - Solved hardware and software bugs using Linux and hardware analysis tools on custom hardware
 - Tools and Technology: C, Python, Linux Kernel, Git, Gerrit
- Herrick Technology Labs** · Software Intern Sep. 2020 to Dec. 2020
- Worked on an inter-disciplinary team tasked with producing a multi-faceted sensing solution for US military avionics
 - Wrote libraries to communicate with HTL radio solutions over a range of protocols including TCP over Ethernet and UART
 - Optimized radio based object tracking algorithm leveraging GPU based hardware acceleration
 - Tools and Technology: C++, Python, Git, Cuda
- Blue Spiral** · Software Developer June 2020 to Aug. 2020
- Wrote iOS application for on-the-spot employee performance reviews using SwiftUI
 - Updated existing image processing pipeline to use the newest Azure OCR API
 - Developed software to detect vegetation levels from drone footage
 - Tools and Technology: C#, Swift, Python
- Ball Bowler** · Software Developer Jan. 2019 to Nov. 2019
- Designed and developed a UI for a miniature bowling lane
 - Implementing scoring logic and user interface on a LattePanda single board computer
 - Captured state of bowling pins using computer vision
 - Tools and Technology: Java, JavaFX
- Blue Spiral** · Software Intern Aug. 2017 to Aug. 2018
- Developed object detection training pipeline built on top of Tensorflow
 - Applied object detection pipeline for the detection of unwanted ducks
 - Incorporated object detection pipeline with the Microsoft HoloLens
 - Tools and Technology: C#, Python, C++, Tensorflow
- Valogix** · Software Intern July 2016 to Nov. 2018
- Implement a system level automated testing suite for Valogix web application
 - Produced custom web API incorporated into Valogix's servers for system health information
 - Developed a web application for keeping track of the over 100 deployed applications
 - Tools and Technology: Java, Spring, Spring Boot, PostgreSQL, Groovy

Activities

Master's Thesis

Augmented Reality for the Visualization of Robotics Control Algorithms

- Developed Unity app for the Meta Quest to enable visualizing RViz2 messages
- Integrate with motion capture to overlay virtual context in the physical world
- Allowed user to view virtual influences acting on the system in real time

Electric Vehicle Team

Firmware Lead

May 2020 to Present

Firmware Member

Dec. 2018 to May 202

- Develop a custom software library for developing firmware on EVT produced hardware (EVT-core)
 - Produce object oriented based software layed on top of the STM32 Hardware Abstraction Layer (HAL)
 - Wrote drivers to support various communication protocols in EVT-core including CAN, UART, I2C, and PWM
- Designed and developed firmware for the EVT produced battery management system
- Added support for CANopen to EVT-core for communication across the motorcycle systems
- Developed drivers for communicating with Sendyne GFD, TI battery management chip, and STM32 peripherals